NEC

NEC Electronics Inc.

MC-41256A8
262,144 x 8-BIT
DYNAMIC NMOS RAM MODULE

Description

The MC-41256A8 is a 262,144-word by 8-bit NMOS RAM module designed to operate from a single +5-volt power supply. Advanced dynamic circuitry, including a single-transistor storage cell, 1024 sense amplifiers per data output, multiplexed address buffers and flexible refresh controls, provides good system operating margins.

The MC-41256A8 is functionally equivalent to eight μPD41256 standard 256K DRAMs. Refreshing is accomplished by means of $\overline{RAS}$-only refresh cycles, hidden refresh cycles, $\overline{CAS}$ before $\overline{RAS}$ refresh cycles, or by normal read or write cycles on the 256 address combinations of A_0-A_7 during a 4-ms period.

Packaged in a Single Inline Memory Module (SIMM™) to enhance reliability and reduce the size, weight and cost of a system, the MC-41256A8 includes eight μPD41256s in PLCC packages and eight power supply decoupling capacitors.

SIMM is a trademark of Wang Laboratories.

Features

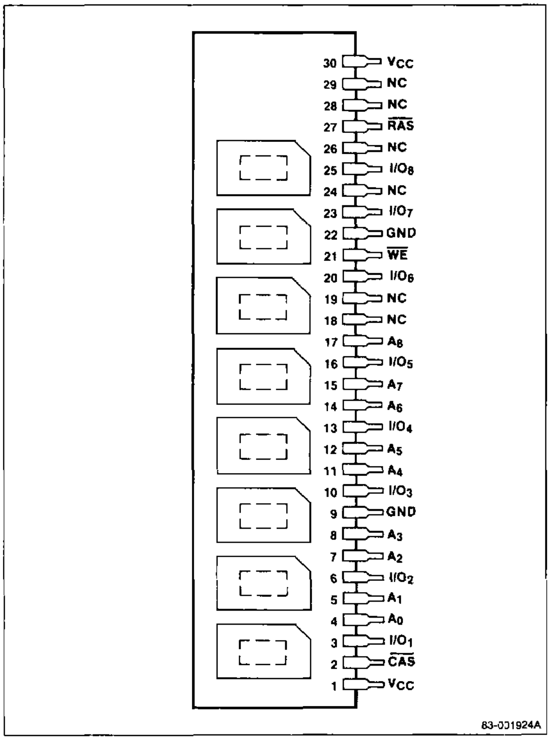
- ☐ 262,144-word by 8-bit organization
- ☐ Single +5-volt ± 10% power supply
- ☐ Standard 30-pin Single Inline Memory Module (SIMM) packaging
- ☐ Eight 256K dynamic RAMs incorporated in high-density PLCC packaging
- ☐ Eight power supply decoupling capacitors included
- ☐ Low power dissipation: 220 mW standby (max)
- ☐ TTL-compatible inputs and outputs
- ☐ 256 refresh cycles (A_0-A_7 are refresh address pins)
- ☐ Page-mode capability

Ordering Information

Part Number	Access Time (max)	Read/Write Cycle Time (min)	Page-Mode Cycle Time (min)	Package
MC-41256A8A-12	120 ns	220 ns	120 ns	30-pin leaded SIMM
A-15	150 ns	260 ns	145 ns	
MC-41256A8B-12	120 ns	220 ns	120 ns	30-pin socketable SIMM
B-15	150 ns	260 ns	145 ns	

Pin Configurations

30-Pin SIMM, MC-41256A8A



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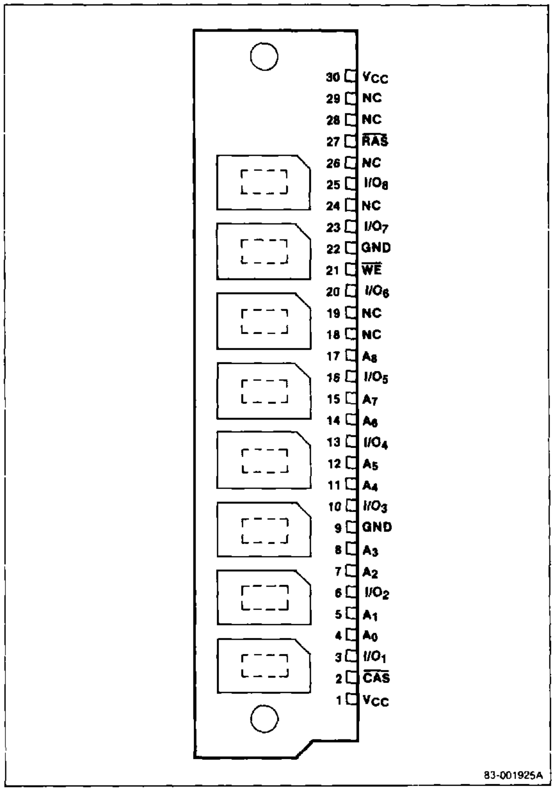
Pin Configurations (cont)

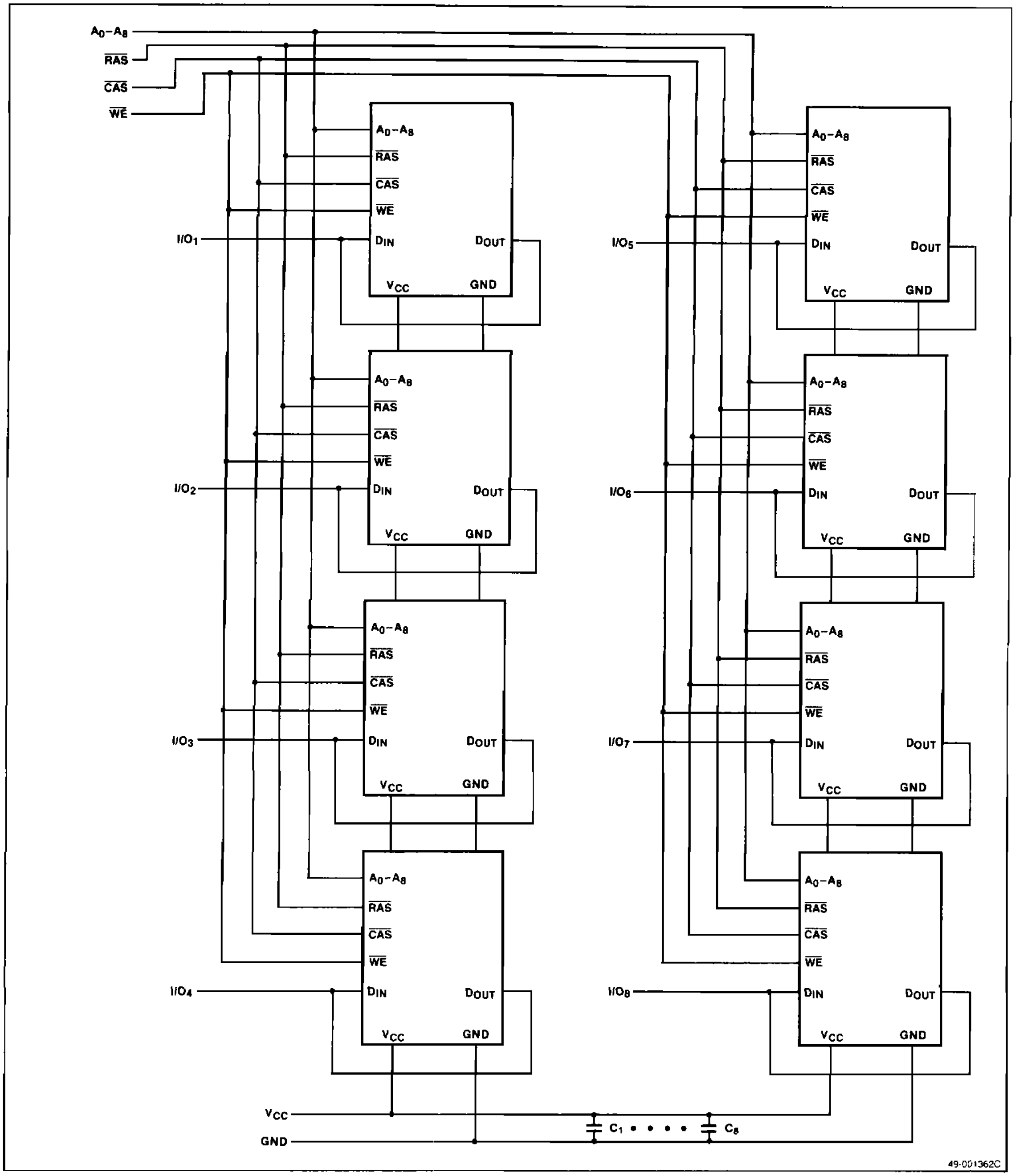
30-Pin SIMM, MC-41256A8B



Pin Identification

Symbol	Function
A_0-A_8	Address inputs
I/O_1-I/O_8	Common data inputs and outputs
$\overline{CAS}$	Column address strobe
$\overline{RAS}$	Row address strobe
$\overline{WE}$	Write enable
GND	Ground
V_{CC}	+5-volt power supply
NC	No connection

Block Diagram



Absolute Maximum Ratings

Voltage on any pin relative to GND	−1.0 to +7.0 V
Operating temperature, T_{OPR}, ambient	0 to +70°C
Storage temperature, T_{STG}	−55 to +125°C
Short-circuit output current, I_{OS}	50 mA
Power dissipation, P_D	8.0 W

Comment: *Exposure to Absolute Maximum Ratings for extended* periods may affect device reliability; exceeding the ratings could cause permanent damage. The device should be operated within the limits specified under DC and AC Characteristics.

Capacitance

T_A = 0 to +70°C; V_{CC} = +5.0 V ±10%; f = 1 MHz

		Limits				
Parameter	**Symbol**	**Min**	**Typ**	**Max**	**Unit**	**Test Conditions**
Input capacitance	C_{IA}			55	pF	A_0-A_8
	C_{IR}			70	pF	$\overline{RAS}$, $\overline{WE}$
	C_{IC}			70	pF	$\overline{CAS}$
Input/output capacitance	C_{DQ}			17	pF	For I/O_1-I/O_8: $\overline{CAS} = V_{IH}$ to disable D_{OUT}

DC Characteristics

T_A = 0 to +70°C; V_{CC} = +5.0 V ±10%; GND = 0 V

		Limits				
Parameter	**Symbol**	**Min**	**Typ**	**Max**	**Unit**	**Test Conditions**
Supply voltage	V_{CC}	4.5	5.0	5.5	V	
Input voltage, high	V_{IH}	2.4		V_{CC} + 1.0	V	
Input voltage, low	V_{IL}	−1.0		0.8	V	
Standby current	I_{CC2}			40.0	mA	$\overline{RAS} = V_{IH}$; D_{OUT} = high-Z
Input leakage current	I_{IL}	−80		80	μA	V_{IN} = 0 to V_{CC}; other pins = 0 V
Output leakage current	I_{OL}	−10		10	μA	D_{OUT} disabled; V_{OUT} = 0 to V_{CC}
Output voltage, low	V_{OL}	0		0.4	V	I_{OUT} = 4.2 mA
Output voltage, high	V_{OH}	2.4		V_{CC}	V	I_{OUT} = −5 mA

AC Characteristics

T_A = 0 to +70°C; V_{CC} = +5.0 V ± 10%

		Limits					
		MC-41256A8-12		MC-41256A8-15			
Parameter	**Symbol**	**Min**	**Max**	**Min**	**Max**	**Unit**	**Test Conditions**
Operating current, average	I_{CC1}		560		480	mA	$\overline{RAS}$, $\overline{CAS}$ cycling; $t_{RC} = t_{RC}$ min (Note 5)
Refresh operating current, average	I_{CC3}		480		400	mA	$\overline{RAS}$ cycling; $\overline{CAS} = V_{IH}$; $t_{RC} = t_{RC}$ min (Note 5)
Page-mode operating current, average	I_{CC4}		400		320	mA	$\overline{RAS} = V_{IL}$; $\overline{CAS}$ cycling; $t_{PC} = t_{PC}$ min (Note 5)
$\overline{CAS}$ before $\overline{RAS}$ refresh operating current, average	I_{CC5}		480		400	mA	$\overline{RAS}$ cycling; $\overline{CAS} = V_{IL}$; $t_{RC} = t_{RC}$ min (Note 5)
Random read or write cycle time	t_{RC}	220		260		ns	(Note 6)
Page-mode cycle time	t_{PC}	120		145		ns	(Note 6)
Refresh period	t_{REF}		4		4	ms	
Access time from $\overline{RAS}$	t_{RAC}		120		150	ns	(Notes 7, 8)
Access time from $\overline{CAS}$	t_{CAC}		60		75	ns	(Notes 7, 9)
Output buffer turnoff delay	t_{OFF}	0	30	0	35	ns	(Note 10)
Rise and fall transition time	t_T	3	50	3	50	ns	(Note 4)
$\overline{RAS}$ precharge time	t_{RP}	90		100		ns	
$\overline{RAS}$ pulse width	t_{RAS}	120	10000	150	10000	ns	
$\overline{RAS}$ hold time	t_{RSH}	60		75		ns	
$\overline{CAS}$ pulse width	t_{CAS}	60	10000	75	10000	ns	
$\overline{CAS}$ hold time	t_{CSH}	120		150		ns	
$\overline{RAS}$ to $\overline{CAS}$ delay time	t_{RCD}	25	60	25	75	ns	(Note 11)
$\overline{CAS}$ to $\overline{RAS}$ precharge time	t_{CRP}	10		10		ns	(Note 12)
$\overline{CAS}$ precharge time (non-page mode)	t_{CPN}	25		25		ns	
$\overline{CAS}$ precharge time (page mode)	t_{CP}	50		60		ns	
$\overline{RAS}$ precharge $\overline{CAS}$ hold time	t_{RPC}	0		0		ns	
Row address setup time	t_{ASR}	0		0		ns	
Row address hold time	t_{RAH}	15		15		ns	
Column address setup time	t_{ASC}	0		0		ns	
Column address hold time	t_{CAH}	20		25		ns	
Column address hold time referenced to $\overline{RAS}$	t_{AR}	80		100		ns	
Read command setup time	t_{RCS}	0		0		ns	
Read command hold time referenced to $\overline{RAS}$	t_{RRH}	20		20		ns	(Note 13)

AC Characteristics (cont)

T_A = 0 to +70°C; V_{CC} = +5.0 V ± 10%

		Limits					
		MC-41256A8-12		MC-41256A8-15			
Parameter	Symbol	Min	Max	Min	Max	Unit	Test Conditions
Read command hold time referenced to $\overline{CAS}$	t_{RCH}	0		0		ns	(Note 13)
Write command hold time	t_{WCH}	30		40		ns	
Write command hold time referenced to $\overline{RAS}$	t_{WCR}	90		115		ns	
Write command pulse width	t_{WP}	20		25		ns	
Write command to $\overline{RAS}$ lead time	t_{RWL}	40		45		ns	
Write command to $\overline{CAS}$ lead time	t_{CWL}	40		45		ns	
Data-in setup time	t_{DS}	0		0		ns	(Note 14)
Data-in hold time	t_{DH}	30		40		ns	(Note 14)
Data-in hold time referenced to $\overline{RAS}$	t_{DHR}	90		115		ns	
Write command setup time	t_{WCS}	0		0		ns	
$\overline{CAS}$ setup time for $\overline{CAS}$ before $\overline{RAS}$ refreshing	t_{CSR}	10		10		ns	(Note 15)
$\overline{CAS}$ hold time for $\overline{CAS}$ before $\overline{RAS}$ refreshing	t_{CHR}	30		30		ns	(Note 15)

Notes:

(1) All voltages are referenced to GND.

(2) An initial pause of 100 μs is required after power-up, followed by any eight $\overline{RAS}$ cycles before proper device operation is achieved.

(3) AC measurements assume t_T = 5 ns.

(4) V_{IH} (min) and V_{IL} (max) are reference levels for measuring timing of input signals. Transition times are measured between V_{IH} and V_{IL}.

(5) I_{CC1}, I_{CC3}, I_{CC4}, and I_{CC5} depend on output loading and cycle rates. Specified values were obtained with the output open.

(6) The minimum specifications are used only to indicate the cycle time at which proper operation over the full temperature range (T_A = 0 to +70°C) is assured.

(7) Load = 2 TTL (−1 mA, +4 mA) loads and 100 pF (V_{OH} = 2.0 V, V_{OL} = 0.8 V).

(8) Assumes that $t_{RCD} \leq t_{RCD}$ (max). If t_{RCD} is greater than the maximum recommended value in this table, t_{RAC} increases by the amount that t_{RCD} exceeds the value shown.

(9) Assumes that $t_{RCD} \geq t_{RCD}$ (max).

(10) t_{OFF} (max) defines the time at which the output achieves the open-circuit condition and is not referenced to V_{OH} or V_{OL}.

(11) Operation within the t_{RCD} (max) limit assures that t_{RAC} (max) can be met. t_{RCD} (max) is specified as a reference point only; if t_{RCD} is greater than t_{RCD} (max), access time is controlled exclusively by t_{CAC}.

(12) The t_{CRP} requirement should be applicable for $\overline{RAS}/\overline{CAS}$ cycles preceded by any cycle.

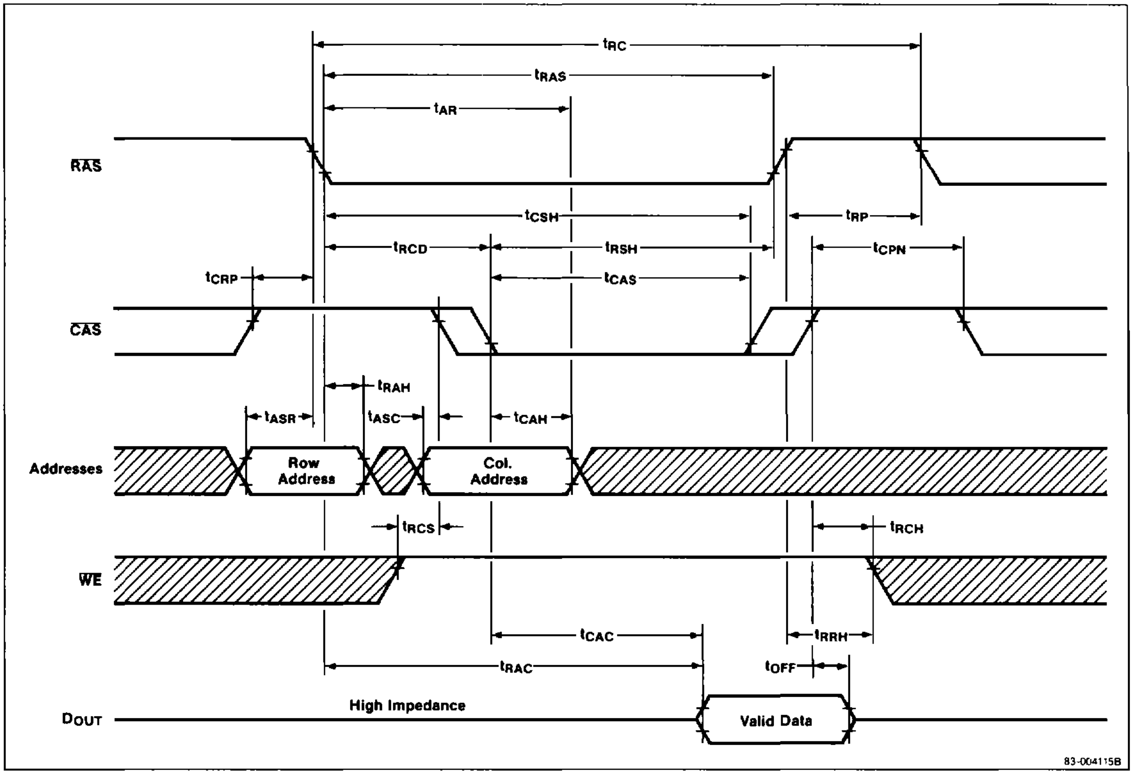
(13) Either t_{RRH} or t_{RCH} must be satisfied for a read cycle.

(14) These parameters are referenced to the leading edge of $\overline{CAS}$.

(15) $\overline{CAS}$ before $\overline{RAS}$ operation is specified.

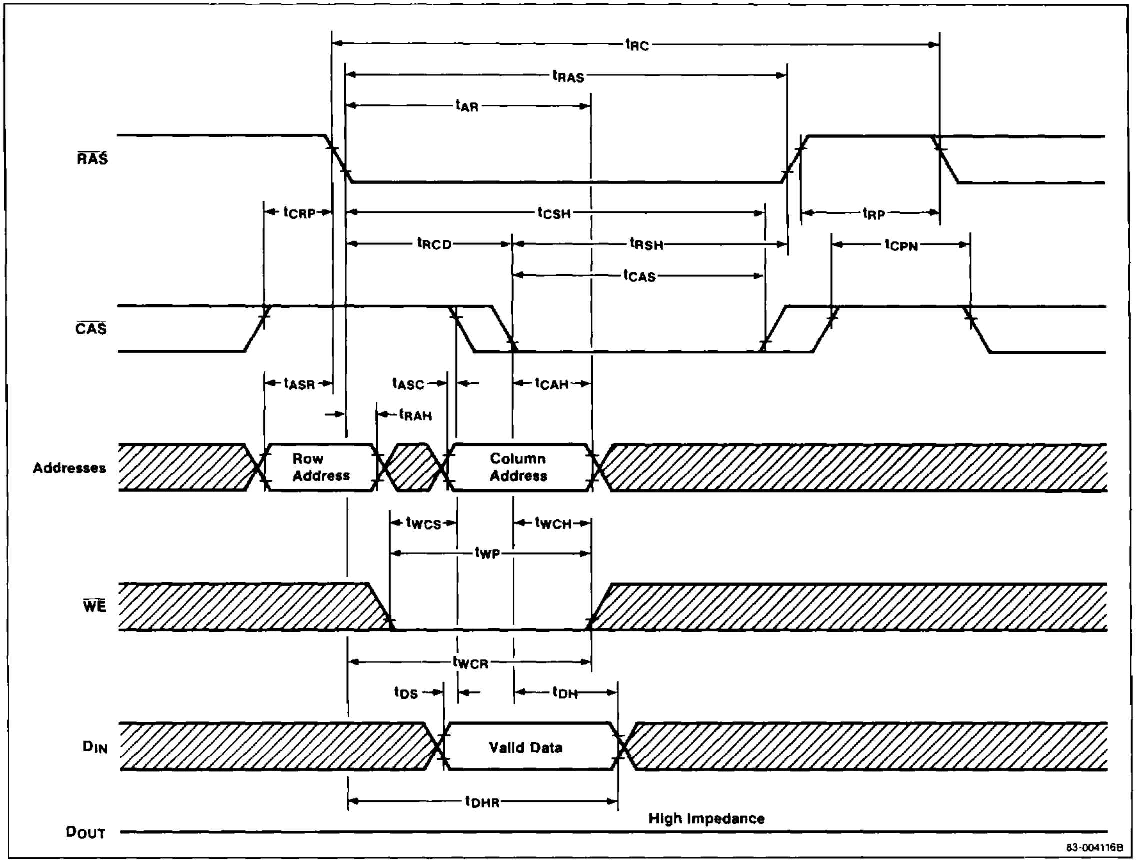
Timing Waveforms

Read Cycle



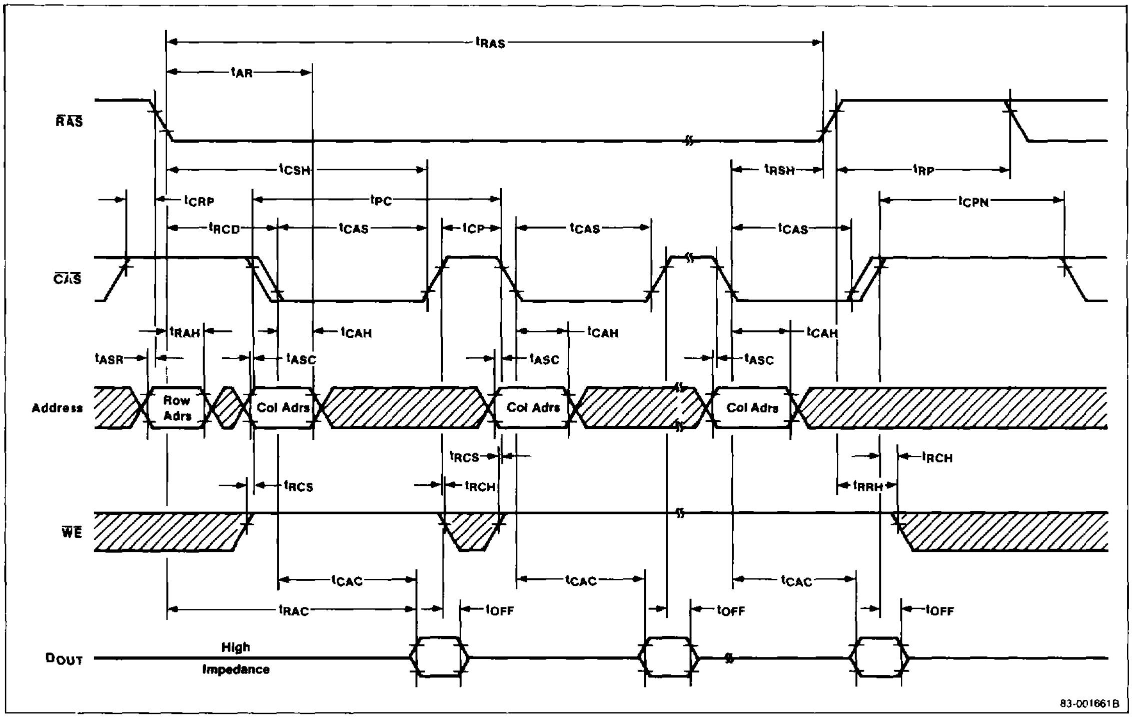
Timing Waveforms (cont)

Write Cycle (Early Write)



Timing Waveforms (cont)

Page Read Cycle



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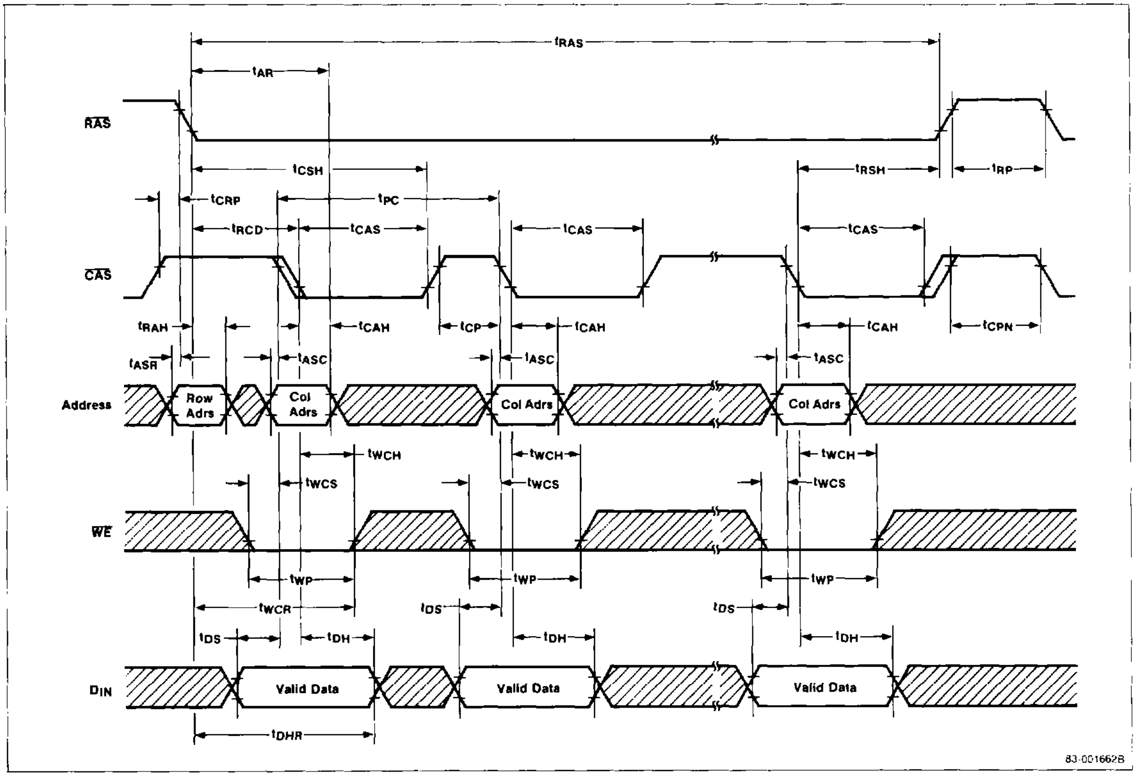
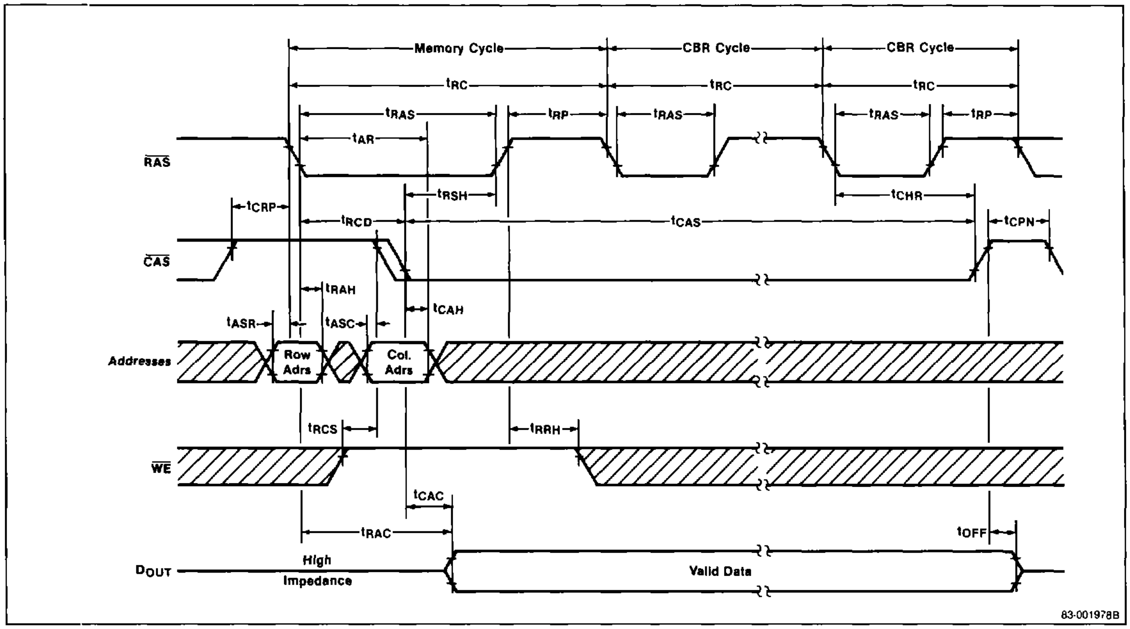
Timing Waveforms (cont)

Page Write Cycle (Early Write)

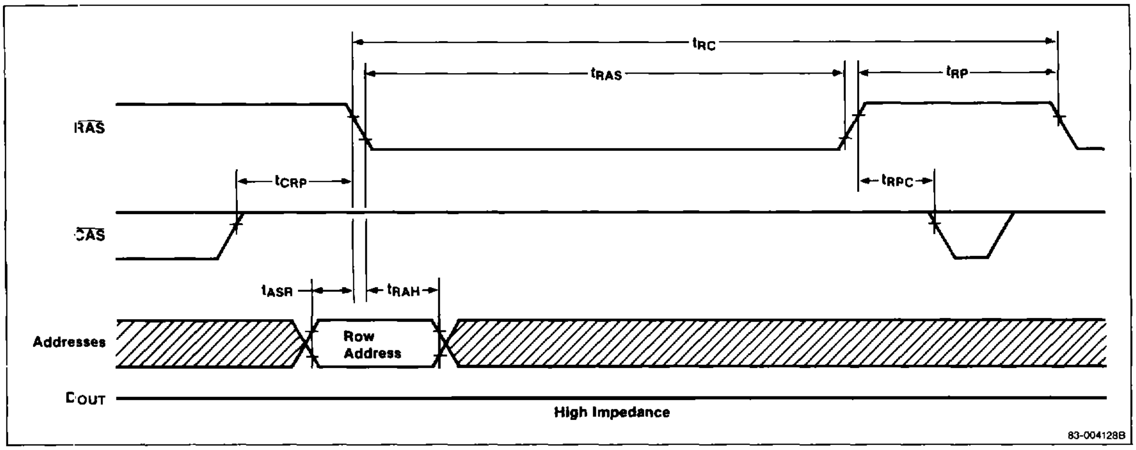


Timing Waveforms (cont)

Hidden Refresh Cycle



RAS-Only Refresh Cycle



Timing Waveforms (cont)

$\overline{CAS}$ Before $\overline{RAS}$ Refresh Cycle